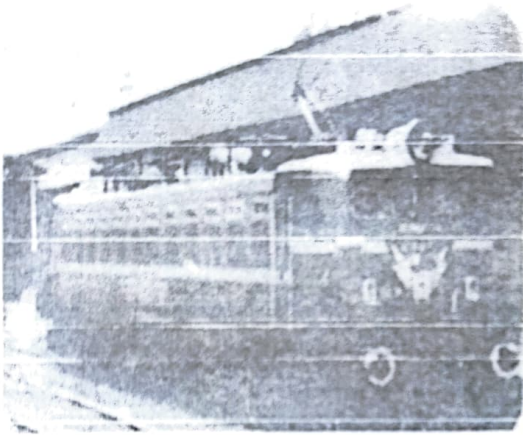


वेस्टर्न रेलवे WESTERN RAILWAY



इलेक्ट्रिक लोको

Electric Loco Shed, VALSAD.

शेड वलसाड

बोगी सेवशन AOH/IOH/TOH चेक शीट

LOCO NO.	:	32458 WAG-9
SCHEDULE	:	TOH.
DATE OF COMMENCEMENT	:	27-10-22
DATE OF COMPLETION	:	09-11-22

SSE/M43R/BL

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TO-H

Date 31/10/22

SN	Schedule Activity	Standard	Actual/Action taken				
I	BOGIE FRAME OVERHAULING (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)						
1	Remove all brake rigging items from the bogie frame before overhaul of bogie frame.(IOH)	Removed	All washers, split pin, cutters changed.				
2	Clean the bogie frame by scrapping.	Cleaned	Buffing done				
3	Make a through visual inspection of bogie frame for cracks, damage and evidence of fatigue.	No crack/ No damage	Visually checked ok				
4	RDPT of TBU hanger bracket tube at weld for any crack. (SMI/0251)WAG9H	No crack	N/A or Conv. B/R				
5	RDPT of the brake hanger and lever to detect any crack.	No crack	RDPT done				
6	Critically check the bogie motor support (lug) for any crack by MPT.	Checked	MPT done				
7	Check the brake rigging for loose, damage or worm out parts. Rectify as required. (TOH)	Checked/ Rectified	checked ok				
8	Replace all pins, bushes and fasteners of brake rigging with new one. (IOH)	Replaced	All washers, split pin, cutters, changed				
9	Replaced the brake blocks with new one.	Replaced	Replaced				
10	Check the brake block key security. Ensure intactness of R-clip and secure with cable tie.	Ensured	N/A or Conv. B/R				
11	Provide safety slings on all TBU + PBU location. (SMI/0244) WAG9H	Provided	N/A or Conv. B/R				
12	Check rubbing of brake hanger with bogie frame.	No rubbing	No rubbing				
	Minimum gap to be maintained between bogie frame & hanger. (SMI/251) WAG9H	As per table below	As per table below				
	Standard	1	2	3	4	5	6
	Gap 13mm	—	—	—	—	—	—
13	Replace the lateral and vertical stops.	Replaced	Replaced				
14	Modification of top bracket for secondary vertical dampers of WAG9 loco to prevent their failure MP.MOD.VL.02.07.09 (Rev.-00)	Replaced	Modified				
15a	Ensure 100% availability of slack adjuster. (SMI/0251) WAG9H	Available	Available				
b	Check length of spring after tightness.	N/A or Conv. B/R					
	Standard	1	2	3	4	5	6
	70 mm	—	—	—	—	—	—
c	Check gap between lug and slack adjuster.						
	Standard	1	2	3	4	5	6
	7.5 mm	—	—	—	—	—	—

BF 1 — Chirag + Dharmesh

BF 2 — Bipin O + Yeshward

Name of Staff

Signature of SE/JE


Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken				
16	Replacement of all sphere blocks.	Replaced					
17	Check the earthing shunts provided on axle box, bogie & body.	Provided	<u>Provided</u>				
18	Maintain the vertical & lateral clearance (Primary & secondary) as per RDSO/TC/082	As per table below	As per table below				
A. Primary vertical clearance (on one bogie)							
	STD	Actual values					
		1	2	3	4	5	6
	27 to 35	<u>30</u>	<u>30</u>	<u>32</u>	<u>32</u>	<u>32</u>	<u>35</u>
B. Primary lateral clearance (on one bogie)							
	STD	Actual values					
		1	2	3	4	5	6
	15 to 22	<u>20</u>	<u>22</u>	<u>18</u>	<u>18</u>	<u>22</u>	<u>22</u>
	AVG						
C. Secondary vertical clearance (on one bogie)							
	STD	1			2		
	32 to 40	<u>38</u>			<u>38</u>		
D. Secondary lateral clearance (on one bogie)							
	STD	1			2		
	45 to 55	<u>50</u>			<u>52</u>		
	AVG						
II	OVERHAUL OF TBU/PBU WAG9H						
1	MPT of brake lever as per SMI 251	Done		<u>7/11/22</u>			
2	Replacement of brake lever if found cracked/deformed	Replaced		<u>Conv. B/R</u>			
3	Replacement of threaded bolts and F.S all steel lock nuts.	Replaced					

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ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date 29/10/22

SN	Schedule Activity	Standard	Actual/Action taken
III	SUSPENSION SPRING		
a	Examine the primary and secondary springs for signs of damage cracks etc.	No damage/ No cracked	No damage & No crack
b	Measure primary and secondary springs characteristic/parameter.	As per table below	Measured

Check Sheet for WAG-9H Secondary Outer Spring
(CLW Drg. No. 1209.01.015.010 Alt.01) IB 011-00842

SN	Test Parameter	STD	Actual			
			1/5	2/6	3/7	4/8
1	Free height	732-735 733.2 mm	732/735	735/735	734/735	734/736
2	Working height	552.0 - 609.5	583/581	581/580	581/581	581/581
3	Spring working load	9871 Kg	—	9871 kg	—	—
4	The end tip should not bite the effective coil	No bite	—	No Bite	—	—
5	None of coil shall be in contact with adjacent coil at working height	No touching	—	No touching	—	—
6	Band 1 positive tolerance Band 2 negative tolerance	575.0 - 609.5 mm 552.0 - 571.0 mm	+ within limit Pro			
7	Spring of same bend should be used in same bogie	Followed	—	Followed	—	—

Check Sheet for WAG-9H Secondary Inner Spring
(CLW Drg. No. 1209.01.015.010 Alt.01) IB 011-00842

SN	Test Parameter	STD	Actual			
			1/5	2/6	3/7	4/8
1	Free height	491 mm	494/497	497/494	494/497	495/497
2	Working height	388.8 - 429.5 mm	405/405	406/405	406/406	408/405
3	Spring working load	1499 Kg	—	1499 kg	—	—
4	The end tip should not bite the effective coil	No bite	—	No bite	—	—
5	None of coil shall be in contact with adjacent coil at working height	No touching	—	No touching	—	—
6	Band 1 positive tolerance Band 2 negative tolerance	405.0 - 429.3 mm 388.8 - 404.0 mm	- Within limit Provided			
7	Spring of same bend should be used in same bogie	Followed	—	Followed	—	—

Sumil C Sutar
Pam Day
Ajay (APP.)

Name of Staff

Signature of SE/JE

Signature of SSE

Loco No. _____ Schedule _____ Date _____

Check Sheet for WAG-9H Primary Outer Spring
(Ref.: CLW Drg. No. IB 011.00843)

SN	Test Parameter	STD	Actual											
			1	2	3	4	5	6	7	8	9	10	11	12
1	Free height	238.8 mm												
2	Working height	184.3-203.5 mm												
3	Spring working load	4140 Kg												
4	The end tip should not bite the effective coil	No bite												
5	None of coil shall be in contact with adjacent coil at working height	No touching												
6	Band 1 positive tolerance Band 2 negative tolerance	192.0-203.5 mm 184.3-191.0 mm												
7	Spring of same bend should be used in same bogie	Followed												

Name of Staff

Signature of SE/JE


Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. _____ Schedule _____ Date _____

Check Sheet for WAG-9H Primary Inner Spring
[Ref.: CLW Drg. No. IB 011.00843]

SN	Test Parameter	STD	Actual											
			1	2	3	4	5	6	7	8	9	10	11	12
1	Free height	240.2 mm												
2	Working height	180.5-199.3 mm												
3	Spring working load	500 Kg												
4	The end tip should not bite the effective coil	No bite												
5	None of coil shall be in contact with adjacent coil at working height	No touching												
6	Band 1 positive tolerance Band 2 negative tolerance	188.0- 199.3 mm 180.5-187.0 mm												
7	Spring of same bend should be used in same bogie	Followed												

Bogie not removed hence spring not checked
Visually checked

[Signature]

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date 02/11/22

SN	Schedule Activity	Standard	Actual/Action taken
III c	Ensure pairing of springs as per band.	Ensured	Ensured
IV	DAMPERS (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
a	Replace all dampers	Replaced	Servicable
b	Replace all FS all steel lock nuts opened for lifting of locos at mentioned location. A. Secondary dampers (Yaw, Vertical, Horizontal) B. Primary dampers top side	Replaced on conditional basis	Replaced
c	Check the alignment of top bracket foundation of damper with bogie frame	Aligned	Checked OK
V	SPHERI BLOCS (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
a	Replace all the Spheri bloc of axle guide, and traction motor torque arm	Replaced	All tm Spheri bloc New for
b	Check intactness of axle guide bolt	Intact	Intact
c	Ensure tightness by specified torque		
	Bolt type	Std torque	Actual
			1 2 3 4 5 6
	Necked down	670 Nm	— Tightness checked ok
	M-24×300	665 Nm	— Tightness checked ok

TM - 1,2,3 - Torq arm - Nik Parichal, Ramesh D, Chirag,
Sugit kr. (App)

TM - 4,5,6 - Torq arm - Jagdish A, Shandilal S, Shish k,
Harsh R

Name of Staff

Signature of SE/JE


Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32 458

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken
VII	MSU (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
1	Greasing to be done in motor suspension unit (MSU) at NDE side	100 gm	Regreasing done
2	Check and tighten all fasteners of NDE side	Checked	checked & tightened
VIII	AXLE (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
1	Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or discoloration.	Done	Done
2	Test the axle for surface and sub-surface defects using a non-destructive technique (Ultrasonic flaw detector). Replace the axle if cracked, defective or damaged.	Good condition	Axle UST done
IX	AXLE BOX (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007 and RDSO's SMI/0246)		
1	Visually inspect the axle boxes for wear or damage. Pay particular attention to the guide rod mounts and limit stops. Replace the axle box if defective.	Good condition	Visually checked OK
2	Open the axle journal box and inspect the condition of lubricating grease, inner races, rollers and cages. Sample the lubricants and analyze for contamination/ metal content.	Done	Checked OK
3	UST of axle racer.	Done	Axle UST done
4	Replace the bearing along with its accessories if worn or damaged. (NEI: RB5080, SKF: CRU150)	Replaced	Serviceable
5	Replace "O" ring of rear cover. (IOH)	Replaced	Replaced
6	Ensure tightness of rear cover bolts (M 16x45) by specified torque. (IOH)	Torque 100Nm	Tightness done
7	Replace the grease of bearing	590 gm	Replaced
8	Tighten the end plate socket head caps screw (M-20x50) with specified torque.	Torque 33mkg	Tightness done
9	Measure the wear of carbon brush and see it does not go below the grooved line and replaces it if required.	Measured/ Replaced	Serviceable
10	Check the intactness of the earth return path on each wheel set.	Done	Intact
11	Replace the axle box front cover 'O' ring.	Replaced	Replaced
12	Ensure tightness of front cover bolts (M 16x45) by specified torque.	Torque 100Nm	Tightness done

Name of Staff _____

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ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

Loco No.		Schedule				
SN	Test	Standard		Wheel Set		
				1	2	3
1	Sr. No. (Axle No.)			TG-36-21/24	TG-62-21	TG-24-24
2	UST of Axle (TC-47)			Done	Done	Done
3	Check wheel thickness	133 + 1 - 0	PE	133.2	133.5	134
			CE	133	133.4	134
4	Check wheel diameter	New : 1097.0 Service Limit : 1016.0	PE	1074.5	1076.5	1076.0
			CE	1073.0	1075.0	1076.0
5	Check root wear	6.0 mm	PE	3.0	2.0	2.5
			CE	3.5	2.5	3.0
6	Check flange wear	3.0 mm	PE	1.0	0.5	0.5
			CE	1.0	0.5	0.5
7	Check trade wear	6.5 mm	PE	2.0	2.0	2.0
			CE	2.0	2.0	2.0
8	Check inside face wheel distance (According to MPMLBD.02.16.01 (Rev-01) Dt:31/1209)	1596 + 1.5 mm - 0.5 mm Service limit : + 3.0 (1599)	1596.11 96.09	1596.40 95.65	1596.22 95.76	
9	<u>Check 'K' value:</u> 107:21 T New limit (317.09 - 317.17)mm - 15 TEETH Condemning limit : 316.494mm			317.10	317.11	317.13

N.K. Panchal

Name of Staff

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ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

SN	Test	Standard		Wheel Set		
				4	5	6
1	Sr. No. (Axle No.)			TG-38-22	TG-43-24	TG-28-22
2	UST of Axle (TC-47)			Done	Done	Done
3	Check wheel thickness	133 + 1 - 0	PE	134	133	133.5
			CE	134	133	133.5
4	Check wheel diameter	New : 1097.0 Service Limit : 1016.0	PE	1072.5	1073.0	1072.0
			CE	1072.5	1072.0	1072.0
5	Check root wear	6.0 mm	PE	3.0	2.0	3.0
			CE	4.0	3.0	4.5
6	Check flange wear	3.0 mm	PE	1.0	0.5	0.5
			CE	1.0	1.0	2.0
7	Check trade wear	6.5 mm	PE	2.5	2.0	2.5
			CE	3.0	2.0	2.5
8	Check inside face wheel distance (According to MPMLBD.02.16.01 (Rev-01) Dt:31/1209)	1596 + 1.5 mm - 0.5 mm Service limit : + 3.0 (1599)		1596.65 95.53	1596.56 95.44	1596.07 95.66
9	Check 'K' value: 107:21 T New limit (317.09 – 317.17)mm - 15 TEETH Condemning limit : 316.494mm			317.10	317.15	317.15

Name of Staff

Signature of SE/JE

Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

Check Sheet for Axle Box Overhauling of WAG-9 Loco
(Ref.: CLW Drg. No. 1209.01.011.088 Alt.01)

SN	Test Parameter	STD	Actual					
			1	2	3	4	5	6
1	Axle box housing	250 + 0.02 mm	- A Box not removed					
	(a) Bore dia	- 0.00 mm						
	(h) Bore length	220 + 0.10 mm						
		- 0.10 mm						
2	Bearing cage condition	Good condition						
3	Bearing roller condition	Good condition						
4	Condition of rear cover	Good condition	Condition OK					

Check Sheet for Axle Box Overhauling of WAG-9 Loco
(Ref.: CLW Drg. No. 1209.01.011.088 Alt.01)

SN	Test Parameter	STD	Actual					
			1	2	3	4	5	6
1	Abutment ring seat dia.	Std. 170 + 0.146 mm - 0.186 mm	- A / Bore not removed					
2	Abutment ring inner dia.	Std. 170 + 0.106 mm - 0.043 mm						
3	Abutment ring interference	0.04 to 0.143 mm						
4	Distance from axle end to abutment ring end	(188.0 to 189.0)mm						
5	Racer seat dia.	150.0 + 0.043 mm - 0.068 mm						
6	Racer inner dia.	(150.0 - 0.025)mm						
7	Racer interference	(0.068 to 0.093)mm						
8	Racer OD before fitment	174.0mm						
9	Racer OD after fitment	--						
10	Distance from axle end to racer end	(22.0 to 23.0)mm						
11	Racer make	NBC/SKF/FAG	FAG/	NBC/	FAG/	NBC/	NBC/	FAG/
			FAG	NBC	FAG	NBC	NBC	FAG

Name of Staff _____

Signature of SF/JE _____

Signature of SSE _____

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken
X	GEAR CASE/ Transmission (RDSO Letter No. EL/3.1.35/16 Dated 30.03.2007)		
1	Open gear case and remove gear oil & cleaning of gear case	Cleaned	Cleaned
2	Check breakage of bolts	Checked	Checked
3	Open oil gauge assembly and replace rubber item and oil gauge glass	Opened Replaced	Done
4	Ensure tightness of magnetic sealing plug & oil filling cap. Apply RTV	Ensured Applied	Ensured & RTV applied
5	Verify the integrity of gauge glass protection cover	Done	done
6	Refill 06 litres new servo syn oil(RR460) in each gear case (RDSO/TC/034 Rev-04)	Filled	Filled
7	Apply RTV at all joints marching faces	Applied	RTV applied
8	Coupled with following bolts with specified torque given below Hex head screw- 16*200 mm Socket head screw- 12*160 mm Socket head cap screw- 12*80 mm Hex head screw- 12*30 mm Hex head screw- 12*70 mm	300(10.9)Nm 205(8.8)Nm 80Nm 80Nm 80Nm	TM 1-6 ALL G/case bolts changed. New Pru
9	Replace all FS all steel lock nuts opened for lifting for locos at torque arm	Replaced	Replaced

Name of Staff

Signature of SF/JE


Signature of SF/JE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

SN	Schedule Activity	Standard	Actual/Action taken
XIII	TRACTION LINK (SMI/0259)		
1	Replace all FS all steel lock nuts opened for lifting of locos at traction link. (SMI/0259)	Replaced	Replaced
2	Check traction rod flange and housing flange against crack by RDPT. (SMI/0259) Traction rod is to be de-bolted completely.	Checked	checked
3	Check centering/ alignment of traction link housing.		
4	Change Aclathen ring. (SMI/0259)	Changed	Servicable
5	Replace flange bolts and nuts of traction bar of size & tightness by specified torque. (SMI/0259) i) M 20 × 65	26.0 Kgm	Replaced & tightness done
6	Use Inter lock washers (Disc lock/ Nord lock) with modified bolt. (SMI/0259)	Used	Used
7	Use F.S all steel lock nuts with modified bolt M20×65. (SMI/0259) This should not be reused.	Used	Used
8	Ensure the intactness of safety cables, pins and R-clips	Ensured	Ensured
9	Check intactness/ security of inner & outer retaining ring, bolt and plate. Ensure tightness at required torque. (80 Nm for outer ring/ 60 Nm for inner ring)	Ensured	Ensured
10	10.9 property class bolts to be replaced with 8.8 property class bolts.	Replaced	Replaced
XIV	TRACTION MOTOR		
i)	Check the condition of traction motor sphere bloc. Replace if worn out, cracked, damaged or showing signs of deterioration to either the rubber or the metal.	Checked	checked TM SP bloc new pro. Adequate Servicable
ii)	Check the suspension tube bearing grease (NDE end) and do the greasing. Grease 100 gm/tube	Checked/ Greasing Done	Greasing done
XV	Primary earthing cable and bracket		
1	Disassemble the locomotive primary earthing system. Inspect and repair, if required, the earthing cable and bracket, earth contact plate, earth brushes, earth brush springs, earth cable glands etc. at the axle box, as prescribed.	Checked/ Replaced	Servicable
2	Check the security of the primary earth fasteners. Tighten the fasteners if necessary	Checked	checked

Name of Staff

Signature of SE/JE


Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date _____

XV. List of must change items of WAG-9 Loco in IOH (Per loco)

SN	Items to be changed	Std. Qty	Actual done	Remark
1	Sphere bloc (Axle guide + Torque arm)	36 Nos.	12	TM of the change
2	Hardware & F.S All steel lock nuts	All	All	
3	'O' Ring of Axle box	24 Nos.	94	
4	'O' Ring of Gear case	12 Nos.	-	Serviceable
5	Gasket of breather & spy glass overhauling	24 Nos.	changed	
6	Traction link hardware (Bolts & Lock nuts)	24 Set	changed	
7	Lateral stop (Lateral & Vertical)	04 Nos.	New Bro	
8	Heli coils	01 Set	-	Serviceable
9	Earth return brush	04 Set	-	Serviceable
10	Aclathen ring	04 Nos.	-	Serviceable
11	Damper	01 Set	-	Serviceable

XVI. Chemical test to be carried out by C & M for WAG-9 Loco

SN	Components	Test	Actual	Remark
1	Main gear	DPT	Visually	checked
2	Bogie lug bracket	DPT	DPT	
3	Traction bar	MPT	MPT	
4	Housing	MPT	MPT	
5	Racer	UST	UST	
6	Brake hanger, lever, tube	DPT	DPT	

PRIMARY AND SECONDARY SUSPENSION LATERAL AND VERTICAL CLEARANCE

Loco No.:

Type of inspection:

Date:

Limits of Clearance	Wheel No./ Axle Box No.											
	1	2	3	4	5	6	7	8	9	10	11	12
Primary vertical -27 to 35												
Primary lateral -15 to 22												
Secondary vertical -32 to 40												
Secondary lateral -45 to 55												

Name of Staff

Signature of SE/JE


Signature of SSE

ELECTRIC LOCO SHED, VALSAD
WAG-9 LOCO (TOH) CHECK SHEET

Loco No. 32458

Schedule TOH

Date 02/11/22

SN	Description	CAB-1	CAB-2
1	CBC No. (Make & Years)	RIL HTFA	RIL HTFA 12-18
Check for any crack by MPT test of CBC assembly items.			
2	A. CBC Coupler body.	checked ok	checked ok
	B. Knuckle.	checked ok	checked ok
	C. Knuckle Pin.	checked ok	checked ok
	D. Clevis.	checked ok	checked ok
	E. Clevis Pin.	checked ok	checked ok
3	Checking of knuckle nose wear & stretch with Go-No Go gauge (Gauge No-3) New - 25.0 mm, Cond.- 9.5 mm	checked ok	checked ok
4	Checking gap between knuckle nose & coupler by gauge no 1 & 2 Gauge 1-135.0 mm Gauge 2-130.0 mm (AOH)	150 gauge not Pass ok	150 gauge not pass ok
5	Check knuckle pin diameter. New - 41.2 mm, Service limit - 38.0 mm	40.5	40.5
6	Check clevis pin diameter. New- 38.0 mm, Service limit- 36.0 mm		
7	Check working of CBC & Anti creep test.	Anti creep ok	Anti creep ok
8	Check condition & proper working of CBC operating handle.	working ok	working ok
9	Check for any crack and damage in yoke & draft gear.	Normal ok	Normal ok
10	Check free lateral movement of CBC shank.	Free	Free
11	Check condition of striker casting liner. New - 8.0mm, Cond.- 2.0 mm	8.0	8.0
12	Check condition of shank wear liner. New- 6.0 mm, Cond.- 1.0 mm	6.0	6.0
13	The diameter of retracing/locking pin & condition Diameter- 16.0 mm	16.0	16.0
14	Hand brake	Modified H/BK O/H & provided	

H/BK - E. Khan & Subject

Name of Staff

Signature of SE/JE

Signature of SSE